Room Number _____

Seat Number _____

Student Number |__|__|__|__|__|__|__|__|__|

ANONYMOUSLY MARKED

(Please DON'T write your name on this exam paper.
Instead, write down your SID clearly.)

THE UNIVERSITY OF SYDNEY

CONFIDENTIAL EXAM PAPER

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Mathematics and Statistics

EXAMINATION

Semester 1 - Final, 2024

MATH2921 Vector Calculus and Differential Eqs (Adv)

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:

This is a CLOSED book exam - no material permitted.

MATERIALS PERMITTED IN THE EXAM VENUE:

(No electronic aids are permitted e.g. calculator, laptops, phones.)

Formula sheet (provided in the exam paper by the Unit Coordinator)

MATERIALS TO BE SUPPLIED TO STUDENTS:

1 x 16-page answer book

Dept-specific answer material - provided by Dept

INSTRUCTIONS TO STUDENTS:

Circle the answers for MCQs (no points for the working).

Show working for ECQs for points.

There are 100 points in all.

Please tick the box to confirm that your examination paper is complete. ☐

For Examiner Use Only

Q	Mark
1	
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Total _____

Multiple Choice Section

There is one and only one correct choice for each question. Please circle the right answer. There are 15 MCQs in all, worth 3 points each for 45 points.

Question 1 **Points: 3**

Let y_1 and y_2 be two solutions of the ODE with the unknown $y(x)$,

$$y'' - y' - 2y = \cos(x).$$

Which of the following is true?

$y = y_1 - y_2$

$y' = y_1' - y_2'$

$y'' = y_1'' - y_2''$

$(y_1'' - y_1' - 2y_1) - (y_2'' - y_2' - 2y_2) = 0$

(A) $y_1 - y_2$ solves $y'' - y' - 2y = \cos(x)$.

(B) $y_1 - y_2$ solves $y'' - y' - 2y = 0$.

(C) $y_1 + y_2$ solves $y'' - y' - 2y = \cos(x)$.

(D) It is possible that $y_1 = 2y_2$.

Question 2 **Points: 3**

Which of the following is true about the type of the ODE with the unknown $y(x)$,

$$y' + y^2 = 1?$$

(A) First order separable ODE.

(B) Linear first order ODE.

(C) Second order separable ODE.

(D) Nonlinear second order ODE.

Question 3 **Points: 3**

Which of the following is true for the Wronskian $W(x)$ of a fundamental set of solutions for the first order linear homogeneous ODE system

$$\vec{y}'(x) = A(x)\vec{y}(x)?$$

(A) $W(x)$ could be equal to 0 for some x.

(B) $W(x)$ must be equal to 0 for some x.

(C) $W(x)$ can't be equal to 0 for any x.

(D) None of the other three is true.

Question 4 **Points: 3**

For (x, y) and (r, θ) as the Cartesian and polar coordinates for $\mathbb{R}^2$ respectively, which of the following is true for a $2D$ region $\mathcal{R}$?

(A) $\iint_{\mathcal{R}} \sqrt{x^2 + y^2} dxdy = \iint_{\mathcal{R}} r^2 drd\theta$

(B) $\iint_{\mathcal{R}} 1 dxdy = \iint_{\mathcal{R}} 1 drd\theta$

(C) $\iint_{\mathcal{R}} \sqrt{x^2 + y^2} dxdy = \iint_{\mathcal{R}} 1 drd\theta$

(D) $\iint_{\mathcal{R}} \sqrt{x^2 + y^2} dxdy = \iint_{\mathcal{R}} r drd\theta$

Question 5 **Points: 3**

For the sphere $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$, the vector field $\overrightarrow{n}(x, y, z) = (x, y, z)$ and the ball $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$, which of the following is true?

(A) $\iiint_B (x + y + z) dV = \iint_S (x, y, z) \cdot \overrightarrow{n} dS$

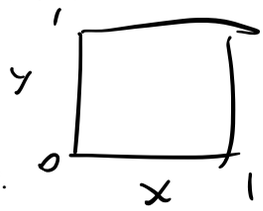
(B) $\iint_S (x, y, 0) \cdot \overrightarrow{n} dS = 0$

(C) $\iiint_B \overrightarrow{n} \cdot \overrightarrow{n} dV = 0$

(D) $\iint_S (y, x^2 - y, zx) \cdot \overrightarrow{n} dS = \iiint_B (x - 1) dV$

Question 6 **Points: 3**

Consider a smooth function $f(x, y)$ over the region $\mathcal{R} = \{(x, y) \mid 0 < x < 1, 0 < y < 1\}$, which of the following is true?



(A) None of the other three is true.

(B) $f(x, y)$ might not have any critical point in $\mathcal{R}$.

(C) $f(x, y)$ always attains maximum and minimum in $\mathcal{R}$.

(D) $f(x, y)$ has at least 1 critical point in $\mathcal{R}$.

Question 7 ✱ **Points: 3**

For a $3D$ region V in $\mathbb{R}^3$ which contains the unit ball centred at the origin, which of the following is always true for the triple integral $\iiint_V 1dV = \iiint_V 1\ dxdydz$?

(A) It is equal to $\iiint_V r\sin\theta drd\theta dz$ in the cylindrical coordinates.

(B) It can be negative.

(C) None of the other three.

(D) It is no bigger than $\iiint_V \rho^2 d\rho d\theta d\varphi$ in the spherical coordinates.

Question 8 **Points: 3**

For the function $f(x,y) = \det\begin{bmatrix} A(x) & B(y) \\ C(x) & D(y)\end{bmatrix} = \begin{vmatrix} A(x) & B(y) \\ C(x) & D(y)\end{vmatrix}$, which of the following is always equal to $f_y(x,y)$?

(A) $\begin{vmatrix} A(x) & B'(y) \\ C(x) & D(y)\end{vmatrix} + \begin{vmatrix} A(x) & B(y) \\ C(x) & D'(y)\end{vmatrix}$

(B) $\begin{vmatrix} 0 & B'(y) \\ 0 & D'(y)\end{vmatrix}$

(C) $\begin{vmatrix} A(x) & B'(y) \\ C(x) & D'(y)\end{vmatrix}$

(D) $\begin{vmatrix} A(x) & B'(y) \\ 0 & D(y)\end{vmatrix} + \begin{vmatrix} 0 & B(y) \\ C(x) & D'(y)\end{vmatrix}$

$f(x,y) = A(x)D(y) - C(x)B(y)$

$f_y = A(x)D'(y) - C(x)B'(y)$

$= \begin{vmatrix} A & B' \\ C & D' \end{vmatrix}$

Question 9 ✱ **Points: 3**

Let $f(x)$ be a continuous function on $[-\pi,\pi]$ with its Fourier series $F(x)$ of period 2π. Which of the following is true?

(A) $F(-\pi)$ might not be equal to $F(\pi)$.

(B) $F(\pi) = f(\pi)$ since $f(x)$ is continuous up to π.

(C) $F(-\pi)$ might not be equal to $f(-\pi)$.

(D) $f(x)$ can always be extended to a continuous function of period 2π over $\mathbb{R}$.

Question 10 **Points: 3**

In $\mathbb{R}^2$, C is the unit circle centred at the origin in the counter-clockwise direction. Which of the following is true?

$\int_0^{2\pi} dt = 2\pi$ $\quad \|\vec{r}(t)\| = 1$

$\vec{r}' = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$

(A) $\int_C ds = \pi$

(B) $\int_{-C} (y, x) \cdot d\vec{r} = \int_C (-y, -x) \cdot d\vec{r}$

(C) $\int_{-C} ds = -2\pi$

(D) $\int_C (x, y) \cdot d\vec{r} < 0$

$-\int_0^{2\pi} \cos^2 t - \sin^2 t \, dt$

$= \int_0^{2\pi} \sin^2 t - \cos^2 t \, dt = \int_0^{2\pi} \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix} \cdot \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} dt$

Question 11 * **Points: 3**

Which of the following is a conservative vector field over the xy-plane $\mathbb{R}^2$?

$\exists \Phi(x,y) : \vec{\nabla} \Phi = \vec{F}$

(A) $(0, x)$

(B) $(-y, x)$

(C) $(y, 0)$

(D) $(0, y)$

Question 12 **Points: 3**

Which of the following is always a normal vector for the level surface $\{(x, y, z) \mid z - f(x, y) = 0\}$ in $\mathbb{R}^3$ at the point $(0, 0, f(0, 0))$?

$\vec{r} = \begin{pmatrix} x \\ y \\ f(x,y) \end{pmatrix}$ $\quad \vec{r}_x = \begin{pmatrix} 1 \\ 0 \\ f_x \end{pmatrix}$ $\quad \vec{r}_y = \begin{pmatrix} 0 \\ 1 \\ f_y \end{pmatrix}$

$\vec{n} = \vec{r}_x \times \vec{r}_y = \begin{pmatrix} -f_x \\ -f_y \\ 1 \end{pmatrix}$

(A) $(-f_y(0, 0), f_x(0, 0), 1)$

(B) $(f_x(0, 0), f_y(0, 0), 1)$

(C) $(f_x(0, 0), f_y(0, 0), -1)$

(D) $(f_x(0, 0), f_y(0, 0), 0)$

Question 13 **Points: 3**

Which pair of functions forms a fundamental set of solutions for the ODE with the unknown $y(x)$,

$$y'' + 3y' + 2y = 0.$$

(A) $x, 2x$.

(B) e^{-x}, e^{-2x}.

(C) $\sin(x), \sin(2x)$.

(D) e^{x}, e^{2x}.

$r^2+3r+2=0$

$(r+2)(r+1)=0$

$r=-1, -2$

Question 14 **Points: 3**

Let $A(x)$ be a 3×3 matrix with entries being functions with the variable x. Which of the following is true?

(A) $e^{A(x)}$ is always invertible for any x.

(B) The determinant of $e^{A(x)}$ might be equal to 0 for some x.

(C) $A(x)e^{A(x)} = e^{A(x)}A(x)$ might not be true for some x.

(D) $\left(e^{A(x)}\right)' = A'(x)e^{A(x)}$ is always true.

$e^{A(x)} = Pe^{D(x)}P^{-1}$

$I = e^{-A(x)}Pe^{D(x)}P^{-1}$

$e^{-A(x)} = Pe^{-D(x)}P^{-1}$

Question 15 **Points: 3**

Which of the following should be the guess for a particular solution of the ODE with the unknown $y(x)$,

$$y'' + 2y' + y = x^2e^{x}$$

(A) $(Ax^2 + Bx + C)e^{x}$

(B) Ax^2e^{x}

(C) $(Ax^2 + Bx + C)e^{-x}$

(D) Axe^{-x}

Extended Answer Questions

Points will be awarded for working. There are 3 questions in all, each with multiple parts for the total of 55 points. Show your working in the space provided after each question. For more space, please use the answer book provided.

Question 16 **Points: 25**

This question is on differential equations.

(a) For the unknown $u(x,t)$ over $\mathbb{R}\times\mathbb{R}$, solve

$$u_{tt} - 16u_{xx} = 0, \quad u(x,0) = e^x, \quad u_t(x,0) = e^{-x}.$$

*(b) For the unknown $u(x,t)$ over $[0,\pi]\times[0,\infty)$, solve

$$u_t - u_{xx} = 0, \quad u(0,t) = u(\pi,t) = 0, \quad u(x,0) = \sin(2x).$$

(c) For the unknown $y(x)$, solve

$y_p = Ax^2e^{-x} + B\cos x + C\sin x$

$r^2+2r+1=0$

$$y'' + 2y' + y = e^{-x} + \sin(x), \quad y(0) = y'(0) = 0.$$

$r=-1$ $y_1 = e^{-x}$ $y_2 = xe^{-x}$

Show your working in the space provided below.

(a) $u_{tt} = 16u_{xx}$, $u(x,0) = e^x$ $u_t(x,0) = e^{-x}$

D'Alemberts Solution:

$$u(x,t) = \frac{1}{2}\left(e^{x-4t} + e^{x+4t}\right) + \frac{1}{2c}\int_{x-4t}^{x+4t} e^{-\tau}\,d\tau$$

$$= e^x\cosh(4t) - \frac{1}{2c}e^{-\tau}\Big|_{x-4t}^{x+4t}$$

$$= e^x\cosh(4t) - \frac{1}{2c}\left(e^{-x-4t} - e^{-x+4t}\right)$$

$$= e^x\cosh(4t) + \frac{e^{-x}}{4}\,\frac{e^{4t}-e^{-4t}}{2}$$

$$= e^x\cosh(4t) + \frac{e^x}{4}\sinh(4t)$$

(b) $c=1$ $L=\pi$, $f(x) = \sin 2x$

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2t}\sin nx$$

$2 - 5 - (2+5)$

$$b_n = \frac{2}{\pi}\int_0^{\pi}\sin 2x \sin nx\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi}\cos(n+2)x - \cos(n-2)x\,dx$$

(c)

Question 17 **Points: 16**

This question is on vector calculus.

(a) Calculate the following integrals. $V = \{(r,\theta,z) \mid 0 \le \theta \le 2\pi,\ 0 \le r \le 2,\ -2 \le z \le 2\}$

(a.1) For $V = \{(x, y, z) \mid x^2 + y^2 \le 4, -2 \le z \le 2\}$ in the Cartesian coordinates,

$$\iiint_V \sin(x^2 + y^2) \cdot (z^2 + 1)\ \underbrace{dxdydz}_{dz\,d\theta\,dr}$$

(a.2) For $R = \{(x, y) \mid x \ge 0, y \le 0, y - x \ge -1\}$ in the Cartesian coordinates,

$$\iint_R xy\ dxdy.$$

$0 \ge y \ge -1$

$0 \le x \le y+1$

$$\int_{-1}^{0}\int_{0}^{y+1} xy\,dx\,dy = \int_{-1}^{0} \frac{x^2 y}{2}\Big|_0^{y+1} dy = \frac{1}{2}\int_{-1}^{0} (y^2+2y+1)\,y\,dy = \frac{1}{2}\int_{-1}^{0} (y^3+2y^2+y)\,dy = \frac{1}{2}\left(\frac{y^4}{4} + \frac{2}{3}y^3 + \frac{y^2}{2}\right)\Big|_{-1}^{0}$$

$$= \frac{1}{2}\left(0 - \frac{1}{4} + \frac{2}{3} - \frac{1}{2}\right) = \frac{1}{2}\left(\frac{2}{3} - \frac{3}{4}\right) = \frac{1}{3} - \frac{3}{8} = \frac{8-9}{24} = -\frac{1}{24}$$

(b) Over the $3D$ region $V = \{(x, y, z) \mid x^2 + y^2 + z^2 \le 1,\ z \ge 0\}$ with the boundary S consisting of the upper hemisphere and the unit disk on the xy-plane, we consider the vector field

$$\vec{F}(x, y, z) = (x^3 + y, y^3 + z, z^3 + x).$$

Applying Gauss' Theorem or otherwise, calculate

$$\iint_S \vec{F} \cdot \vec{n}\ dS$$

where $\vec{n}$ is the unit normal vector field of S pointing in the direction of leaving the region V.

Show your working in the space provided below.

(b) $\vec{F} = \begin{pmatrix} x^3+y \\ y^3+z \\ z^3+x \end{pmatrix}$

$S = \{(x,y,z) \mid x^2+y^2+z^2 = 1,\ z \ge 0\}$

$$\iint_S \vec{F}\cdot\vec{n}\,dS = \iiint_V \vec{\nabla}\cdot\vec{F}\,dx\,dy\,dz$$

$x = r\cos\theta\sin\phi$

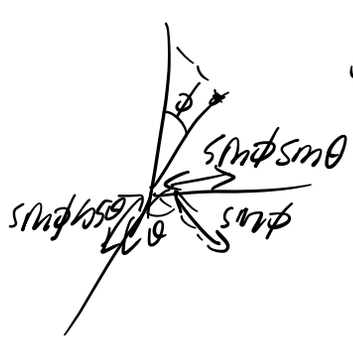
$y = r\sin\theta\sin\phi$

$z = r\cos\phi$

$$= \iiint_V (3x^2+3y^2+3z^2)\,dx\,dy\,dz$$

$$= 3\iiint_V (x^2+y^2+z^2)\,dV$$

$V = \{(r,\phi,\theta) \mid \phi \in [0,\tfrac{\pi}{2}],\ \theta \in [0,2\pi],\ r \in [0,1]\}$



$$\iint_S \vec{F}\cdot\vec{n}\,dS = 3\iiint_V (\sin^2\phi + \cos^2\phi)\,r^2\sin\phi\,dV$$

$$= 3\int_0^{\frac{\pi}{2}}\int_0^{2\pi}\int_0^1 r^4\sin\phi\,dr\,d\theta\,d\phi$$

$$= \frac{3}{5}\int_0^{\frac{\pi}{2}}\int_0^{2\pi} \sin\phi\,d\theta\,d\phi$$

$$= \frac{6\pi}{5}\int_0^{\frac{\pi}{2}} \sin\phi\,d\phi = -\frac{6\pi}{5}\cos\phi\Big|_0^{\frac{\pi}{2}}$$

turn to page 15

$= \frac{6\pi}{5}$

Question 18 **Points: 14**

This question is comprehensive.

(a) Over the xy-plane $\mathbb{R}^2$, let $\Phi(x, y) = x^2 + 2xy + y^2$. Consider the curve $\vec{r}(t) = \big(x(t), y(t)\big)$ satisfying

$\vec{\nabla}\Phi = \begin{pmatrix} 2y+2x \\ 2x+2y \end{pmatrix}$

$$\vec{r}\,'(t) = \nabla\Phi\big(\vec{r}(t)\big).$$

where the right hand side of the equation is the (vector) value for the gradient vector field $\nabla\Phi$ at the point $\vec{r}(t)$.

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

(a.1) Write down the 1st order ODE system for the unknown $\vec{r}(t)$. You might write down $\vec{r}(t)$ in the column form $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ instead of the row form.

$A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

(a.2) Solve the first order ODE system from (a.1), and get the curve $\vec{r}(t)$ starting from $(0, 1)$, in other words, $\vec{r}(0) = (0, 1)$.

$\vec{r}(t) = e^{tA}\vec{r}(0)$

(a.3) For the curve C from (a.2) for $t \in [0, 1]$, calculate

$$\int_C \nabla\Phi \cdot d\vec{r}.$$

$(2-\lambda)^2 - 4 = 0$

$\lambda^2 - 4\lambda = 0$

$\lambda(\lambda - 4) = 0$

$\lambda = 0, 4$

(b) For $x > 0$ and $y > 0$, solve the first order ODE

$$\frac{dy}{dx} = -\frac{x^2 + y^2}{xy}$$

by the following steps.

(b.1) Rewrite the ODE (formally) as $(x^2 + y^2)dx + xydy = 0$. Show that the differential form $(x^2 + y^2)dx + xydy$ is not exact, in other words, the vector field $\vec{F}(x, y) = (x^2 + y^2, xy)$ is not a gradient vector field.

(b.2) Multiply $\vec{F}(x, y) = (x^2 + y^2, xy)$ by an unknown function $\mu(x)$ (only depending on x). Solve the differential equation

$$[\mu(x)(x^2 + y^2)]_y - [\mu(x)xy]_x = 0$$

for $\mu(x)$.

(b.3) Show that the vector field $\mu(x)\vec{F}(x, y)$ from (b.2) is conservative, or equivalently, $\mu(x)\vec{F}(x, y)$ is a gradient vector field, $\nabla\Psi(x, y)$. (Remark: finally, $\Psi(x, y) = C$ for proper constant C solves the original ODE implicitly.)

Show your working in the space provided below.

Formula Sheet

Formulas and theorems provided below might be stated without the complete conditions under which they apply. The notations used are the same as those used in lectures.

Line Integrals

$$\int_{\mathcal{C}} \phi(x,y,z)\,ds = \int_a^b \phi(x(t),y(t),z(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt$$

$$\int_{\mathcal{C}} \overrightarrow{F} \cdot d\overrightarrow{r} = \int_{\mathcal{C}} F_1\,dx + F_2\,dy + F_3\,dz = \text{Work done by } \overrightarrow{F} \text{ along } \mathcal{C}$$

Gradient (in $\mathbb{R}^3$ for example)

$$\text{grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\overrightarrow{i} + \frac{\partial\phi}{\partial y}\overrightarrow{j} + \frac{\partial\phi}{\partial z}\overrightarrow{k}$$

If $\overrightarrow{F} = \nabla\phi$ for a smooth function ϕ (in other words, a gradient vector field), then equivalently $\overrightarrow{F}$ is a conservative vector field (with ϕ being a potential function of $\overrightarrow{F}$), or $\int_{\mathcal{C}} \overrightarrow{F} \cdot d\overrightarrow{r}$ is path-independent.

Curl (for $\overrightarrow{F} = (F_1, F_2, F_3)$)

$$\text{curl } \overrightarrow{F} = \nabla \times \overrightarrow{F} = \begin{vmatrix} \overrightarrow{i} & \overrightarrow{j} & \overrightarrow{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

If the domain of $\overrightarrow{F}$ has no hole and $\nabla \times \overrightarrow{F} = \overrightarrow{0}$, then $\overrightarrow{F}$ is conservative.

Divergence: $\text{div}\,\overrightarrow{F} = \nabla \cdot \overrightarrow{F}$

Double Integral

$$Area(\mathcal{R}) = \iint_{\mathcal{R}} dxdy$$

Signed volume under the graph of $z = f(x,y)$ over $\mathcal{R}$ is

$$\iint_{\mathcal{R}} f(x,y)\,dxdy$$

In polar coordinates,

$$\iint_{\mathcal{R}} f(x,y)\,dxdy = \iint_{\mathcal{R}} f(r\cos\theta, r\sin\theta)\;rdrd\theta$$

Triple Integral

In *Cartesian coordinates* (x, y, z):

$$\iiint_V f(x,y,z)\,dV = \iiint_V f(x,y,z)\,dxdydz.$$

In *cylindrical coordinates* (r, θ, z):

$$\iiint_V f(x,y,z)dV = \iiint_V f(r\cos\theta, r\sin\theta, z)\,rdrd\theta dz.$$

In *spherical coordinates* (ρ, θ, φ):

$$\iiint_V f(x,y,z)\,dV = \iiint_V f(\rho\cos\theta\sin\varphi, \rho\sin\theta\sin\varphi, \rho\cos\varphi)\,\rho^2\sin\varphi d\rho d\theta d\varphi$$

Green's Theorem/Formula (in $\mathbb{R}^2$)

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot d\overrightarrow{r} = \oint_{\mathcal{C}} F_1\,dx + F_2\,dy = \iint_{\mathcal{R}} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dxdy$$

with $\mathcal{C}$ bounding $\mathcal{R}$ in the counter-clockwise way. Green's Theorem is equivalent to **Divergence Theorem in 2D**

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot\overrightarrow{n}\,ds = \iint_{\mathcal{R}} \nabla\cdot\overrightarrow{F}\,dxdy$$

Surface Integral (for the graph S of $z = f(x, y)$ over $\mathcal{R}$)

$$\iint_S \phi(x,y,z)\,dS = \iint_{\mathcal{R}} \phi(x,y,f(x,y))\sqrt{1+f_x^2+f_y^2}\,dxdy$$

In a general coordinate $\overrightarrow{T}(u, v)$, $\sqrt{1+f_x^2+f_y^2}dxdy$ is replaced by $\|\overrightarrow{T}_u \times \overrightarrow{T}_v\|dudv$.

Flux of $\overrightarrow{F}$ through S in the direction of (the unit normal) $\overrightarrow{n}$ is

$$\iint_S \overrightarrow{F}\cdot\overrightarrow{n}\,dS.$$

Gauss' Theorem (also known as Divergence Theorem in 3D)

$$\iint_S \overrightarrow{F}\cdot\overrightarrow{n}\,dS = \iiint_V \nabla\cdot\overrightarrow{F}\,dV$$

for the unit outward normal $\overrightarrow{n}$ of the surface S which the boundary of the $3D$ region V.

Stokes' Theorem

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot d\overrightarrow{r} = \iint_S (\nabla\times\overrightarrow{F})\cdot\overrightarrow{n}\,dS$$

turn to page 3

where $\overrightarrow{n}$, $\mathcal{C}$ and S are compatible.

Abel's Identity/Formula

For the ODE over $[a, b]$,

$$Ly = y'' + p(x)y' + q(x)y = 0,$$

the Wronskian of 2 solutions $W(y_1, y_2)$ satisfy that for any $x_0 \in [a, b]$,

$$W' = -p(x)W, \quad W(x) = W(x_0)e^{-\int_{x_0}^{x} p(t)dt}.$$

For the $n \times n$ ODE system over $[a, b]$,

$$\overrightarrow{y}'(x) = A(x)\overrightarrow{y}(x),$$

the Wronskian of n solutions $W(\overrightarrow{y}_1, \cdots, \overrightarrow{y}_n)$ satisfies for any $x_0 \in [a, b]$,

$$W' = \mathrm{Tr}\big(A(x)\big)W, \quad W(x) = W(x_0)e^{\int_{x_0}^{x} \mathrm{Tr}\big(A(t)\big)dt}.$$

Variation of Constants

For the first order linear ODE system

$$\overrightarrow{y}'(x) = A(x)\overrightarrow{y}(x) + \overrightarrow{f}(x),$$

given the Fundamental Solution Matrix $\Phi(x)$ for the corresponding homogeneous system, the general solution is

$$\overrightarrow{y}(x) = \Phi(x)\int_{x_0}^{x} \Phi^{-1}(t)\overrightarrow{f}(t)dt + \Phi(x)\overrightarrow{C}$$

for any constant vector $\overrightarrow{C}$.

Variation of Parameters

For the second order linear ODE,

$$y'' + p(x)y' + q(x)y = f(x),$$

given the fundamental set of the corresponding homogeneous ("complementary") ODE $\{y_1, y_2\}$, the general solution is

$$y = u_1(x)y_1(x) + u_2(x)y_2(x), \quad u_1 = -\int \frac{y_2 \cdot h}{W(y_1, y_2)}\, dx, \quad u_2 = \int \frac{y_1 \cdot h}{W(y_1, y_2)}\, dx.$$

turn to page 4

Fourier Series

For a function f defined over $[-L, L]$ for some $L > 0$, the *Fourier series* is:

$$\mathcal{F}(f)(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos\left(\tfrac{n\pi}{L}x\right) + b_n \sin\left(\tfrac{n\pi}{L}x\right)]$$

where the *Fourier coefficients* b_0, $(b_n)_{n\geq 1}$ and $(a_n)_{n\geq 1}$ are given by

$$\begin{aligned} a_0 &= \tfrac{1}{2L}\int_{-L}^{L} f(x)\,dx \\ a_n &= \tfrac{1}{L}\int_{-L}^{L} f(x)\cos\left(\tfrac{n\pi}{L}x\right)\,dx \qquad \text{for every } n \geq 1, \\ b_n &= \tfrac{1}{L}\int_{-L}^{L} f(x)\sin\left(\tfrac{n\pi}{L}x\right)\,dx \qquad \text{for every } n \geq 1. \end{aligned}$$

For $y'' + \lambda y = 0$, we have the following **Eigenvalue Problems**:

- Eigenvalue Problem (a) on $[0, L]$: $y(0) = y(L) = 0$; eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunction $\sin(\frac{k\pi}{L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (b) on $[0, L]$: $y'(0) = y'(L) = 0$; eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunction $\cos(\frac{k\pi}{L}x)$ for $k = 0, 1, 2, \cdots$
- Eigenvalue Problem (c) on $[0, L]$: $y'(0) = y(L) = 0$; eigenvalue $\lambda = \frac{(2k-1)^2\pi^2}{4L^2}$ with eigenfunction $\cos(\frac{(2k-1)\pi}{2L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (d) on $[0, L]$: $y(0) = y'(L) = 0$; eigenvalue $\lambda = \frac{(2k-1)^2\pi^2}{4L^2}$ with eigenfunction $\sin(\frac{(2k-1)\pi}{2L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (e) on $[-L, L]$: $y(-L) = y(L) = 0, y'(-L) = y'(L)$; eigenvalue 0 for eigenfunction 1, eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunctions $\cos(\frac{k\pi}{L}x)$ and $\sin(\frac{k\pi}{L}x)$ for $k = 1, 2, \cdots$

For $u_{tt} - C^2 u_{xx} = 0$ with constant C for $(x, y) \in \mathbb{R}^2$ and the initial conditions, $u(x, 0) = f(x), u_t(x, 0) = g(x)$, the **D'Alembert's Solution** is

$$u(x,t) = \frac{1}{2}[f(x+Ct) + f(x-Ct)] + \frac{1}{2C}\int_{x-Ct}^{x+Ct} g(s)ds.$$

turn to page 5

Table of Standard Integrals

1. $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

2. $\int \frac{dx}{x} = \ln|x| + C$

3. $\int e^x \, dx = e^x + C$

4. $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$

5. $\int \sin x \, dx = -\cos x + C$

6. $\int \cos x \, dx = \sin x + C$

7. $\int \sec^2 x \, dx = \tan x + C$

8. $\int \sinh x \, dx = \cosh x + C$

9. $\int \cosh x \, dx = \sinh x + C$

10. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a} + C$

11. $\int x \, \sin x \, dx = -x \cos x + \sin x + C$

12. $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

13. $\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1} \frac{x}{a} + C = \ln\left(x + \sqrt{x^2+a^2}\right) + C'$

14. $$\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1} \frac{x}{a} + C \quad (x > a)$$
$$= \ln\left|x + \sqrt{x^2-a^2}\right| + C' \quad (x > a \text{ or } x < -a)$$

This is the end of the exam paper.